

ENVS 193DS Homework 3

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Homework Setup

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
# creating new object from data set
kelp <- read_csv(here("data", "temp-kelp.csv"))
bike <- read_csv(here("data", "ENVS-193DS-Personal-Data-Project-cont.csv"))
```

Problem 1. Giant kelp fronds

a. An appropriate test

The two tests that are appropriate to describe the strength of the relationship between temperature and giant kelp frond elongation rate are a Pearson's correlation test and a Spearman rank correlation test. These correlation tests can tell us whether these two variables are related to each other. Given that we do not have a specific hypothesis on how temperature affects kelp frond elongation rate, both tests should be used to accurately describe the relationship.

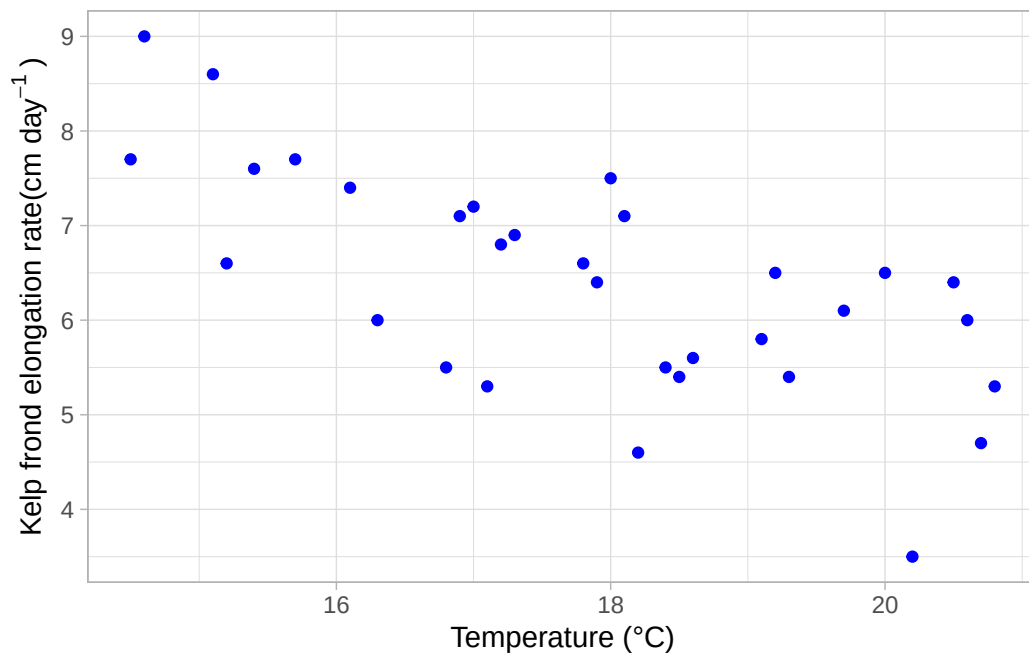
b. Create a visualization

```
# base layer: ggplot
ggplot(data = kelp, # data frame: kelp
       # x-axis: temperature
       # y-axis: kelp frond elongation rate
```

```

mapping = aes(x = temp_c,
              y = kelp_elong)) +
# first layer: points to represent individual observations
geom_point(color = "blue") + # using custom colors
# relabeling axes, adding title
labs(x = "Temperature (°C)",
     y = expression("Kelp frond elongation rate(cm day"-1~")")) +
# changing theme
theme_light()

```



c. Check your assumptions and run your test

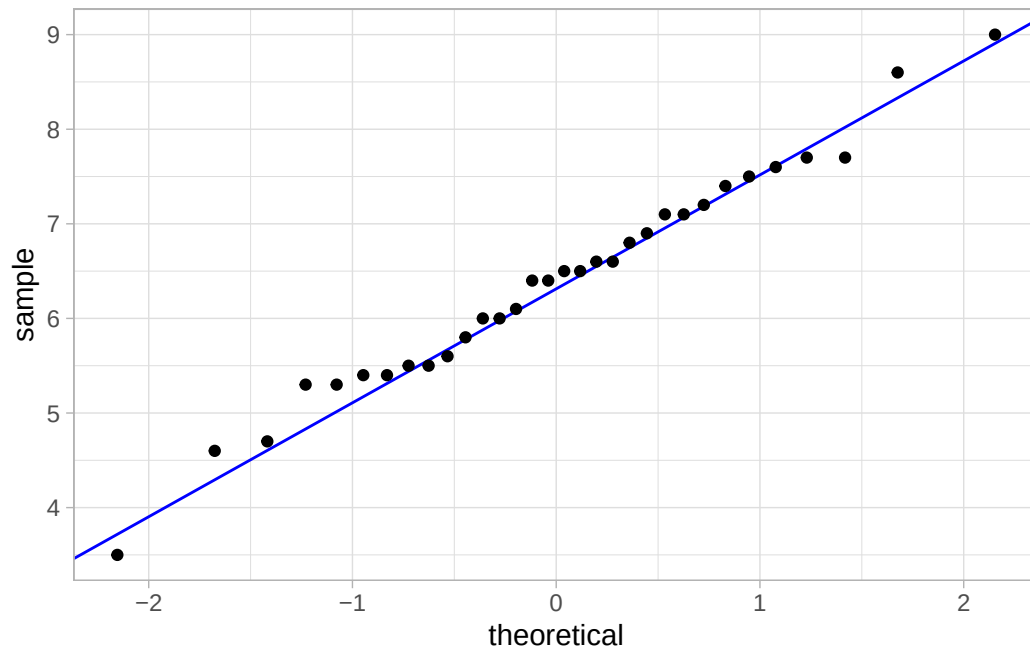
Checking assumptions

```

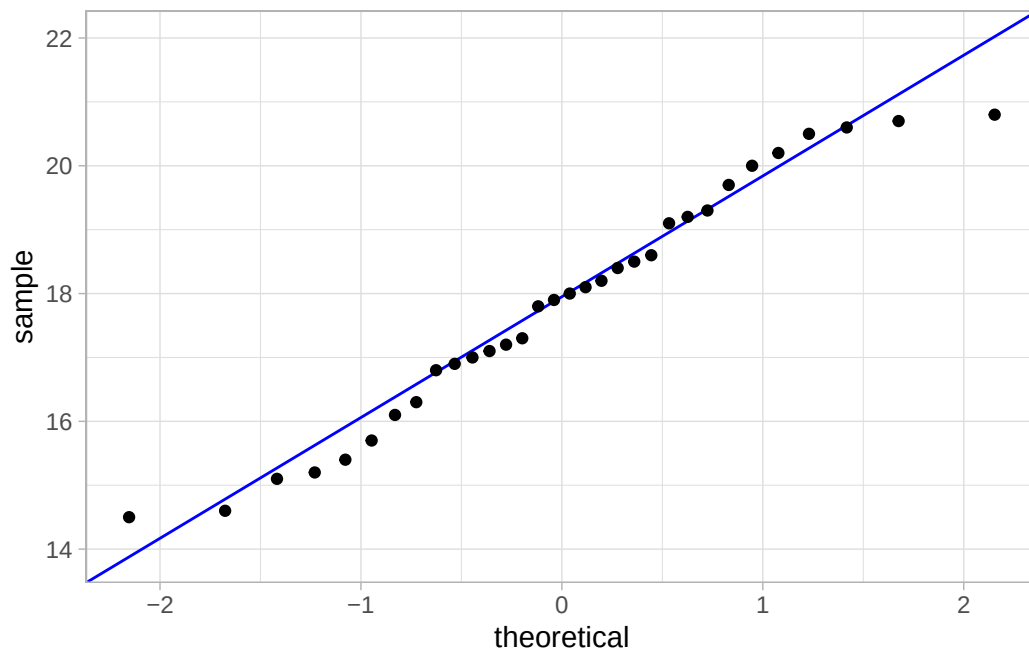
# base layer: ggplot
ggplot(data = kelp, # data frame: kelp
       # y-axis for QQ plot (no x-axis for QQ plot)
       aes(sample = kelp_elong)) +
# first layer: QQ reference line
geom_qq_line(color = "blue") +

```

```
# second layer: QQ plot
geom_qq() +
# custom theme
theme_light()
```



```
# base layer: ggplot
ggplot(data = kelp, # data frame: kelp
       # y-axis for QQ plot (no x-axis for QQ plot)
       aes(sample = temp_c)) +
# first layer: QQ reference line
geom_qq_line(color = "blue") +
# second layer: QQ plot
geom_qq() +
# custom theme
theme_light()
```



Based on the initial visualization, we can expect a linear relationship between temperature and kelp frond elongation rate for Pearson's correlation and a monotonic relationship between the variables for Spearman rank correlation. Both temperature and kelp frond elongation rate are continuous variables. Both variables are normally distributed, based on points roughly following the reference line on the QQ plot. Finally, observations are assumed to be independent.

Running test

```
cor.test(kelp$kelp_elong, kelp$temp_c, # variables: kelp elongation rate and temperature
         method = "pearson") # method: pearson's correlation (r)
```

Pearson's product-moment correlation

```
data: kelp$kelp_elong and kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
```



```
cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, we used a Pearson's correlation (r) test. A correlation test was appropriate for this as we wanted to determine the strength of the relationship between these two variables.

We found a moderate negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's $r = -0.69$, $t(30) = -5.12$, $p < 0.001$, $\alpha = 0.05$).

e. Test implications

From this study, we found that as ocean temperature increases, giant kelp frond elongation rate decreases. This means that, in order to protect and help the growth of giant kelp forests, more research and action should be taken to curb global warming of oceans. Preventing the warming of oceans will directly help giant kelp forests continue to grow.

f. Double check your work

```
cor.test(kelp$kelp_elong, kelp$temp_c, # variables: kelp elongation rate and temperature
         method = "spearman") # method: spearman rank correlation (rho)
```

Spearman's rank correlation rho

```
data: kelp$kelp_elong and kelp$temp_c
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

We found a moderate negative relationship between ocean temperature and giant kelp frond elongation rate (Spearman $\rho = -0.69$, $S = 9216$, $p < 0.001$, $\alpha = 0.05$).

Both Pearson's and Spearman rank correlation test results would have led me to the same conclusion: as ocean temperature increases, giant kelp frond elongation rate seems to decrease.

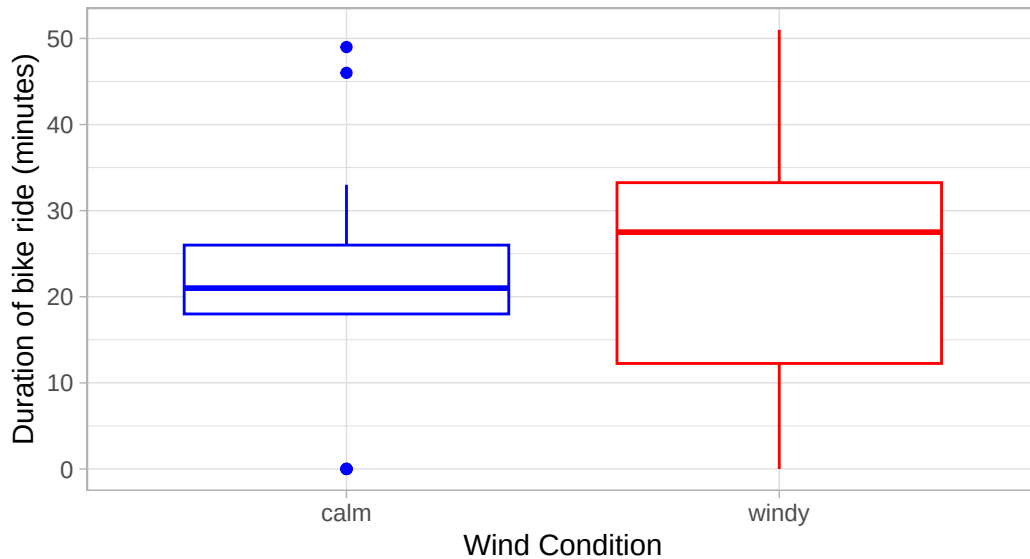
Problem 2. Personal Data

a. Updating your visualizations

```
# base layer: ggplot
ggplot(data = bike, # data frame: bike
      # x-axis: weather
      # y-axis: Duration of bike ride
      # coloring by weather
      aes(x = Weather,
          y = `Duration of bike ride (minutes)`,
          color = Weather)) +
# first layer: boxplot
geom_boxplot() +
# relabeling the axes, adding title and subtitle
labs(x = "Wind Condition",
     y = "Duration of bike ride (minutes)",
     title = "Biking in windy conditions increases the time spend biking",
     subtitle = "Most recent observation: 5/23/2026") +
# custom theme
theme_light() +
# removing legend
theme(legend.position = "none") +
# changing the color
scale_color_manual(values = c("calm" = "blue",
                              "windy" = "red"))
```

Biking in windy conditions increases the time spend biking

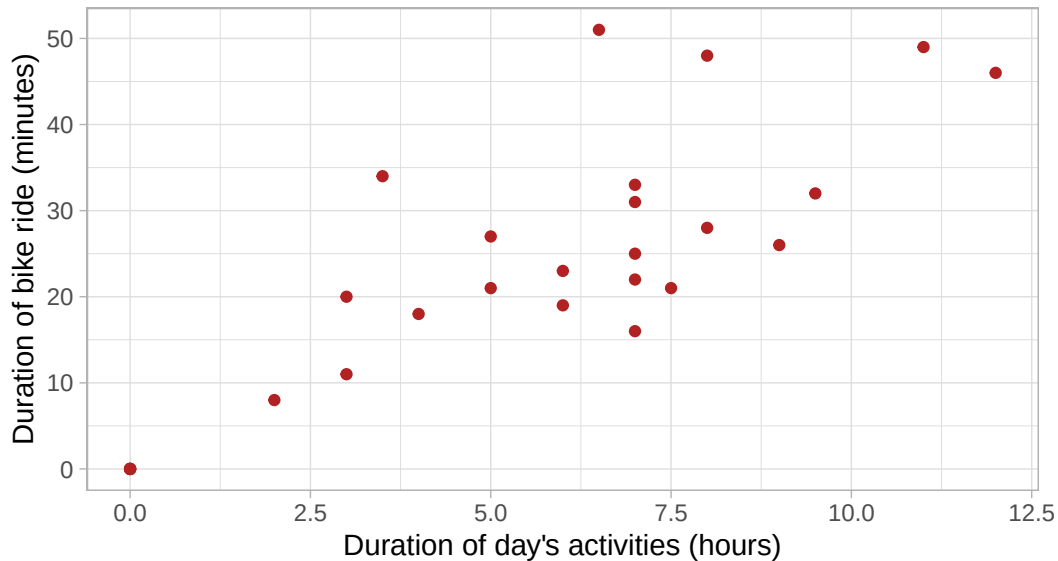
Most recent observation: 5/23/2026



```
# base layer: ggplot
ggplot(data = bike, # data frame: bike
       # x-axis: duration of day's activities
       # y-axis: Duration of bike ride
       aes(x = `Duration of day's activities (hours)`,
           y = `Duration of bike ride (minutes)`),) +
  # first layer: points
  geom_point(color = "firebrick") +
  # relabeling the axes, adding title and subtitle
  labs(x = "Duration of day's activities (hours)",
       y = "Duration of bike ride (minutes)",
       title = "Daily Duration of Activities Effect on Time Spent Biking",
       subtitle = "Most recent observation: 5/23/2026") +
  # custom theme
  theme_light() +
  # removing legend
  theme(legend.position = "none")
```

Daily Duration of Activities Effect on Time Spent Biking

Most recent observation: 5/23/2026



b. Captions

Figure 1. Biking in windy conditions increases the time spend biking Difference in time spent biking between calm and windy conditions is shown as a boxplot. Median, IQR, errorbars, and outliers are visualized and wind condition type is differentiated based on color. Wind data citation: Weather Underground SBFD Station 1 - KCASANTA4905

Figure 2. Daily Duration of Activities Effect on Time Spent Biking As duration of day's activities increases, time spent biking also increases. Individual daily observations are shown as dots on the graph.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data.

I think an interesting affective visualization for my personal data could be a sort of bike race to a top of a mountain. This would be for my scatterplot, where duration of daily total bike rides is visualized as a function of the duration of day's activities. Each point could represent a ridge line or outcropping and the slope of the mountain would roughly follow the data points

upward. Given that this is graph about my bike riding tendencies, I think this would be a creative and interesting way to represent that.

b. Create a sketch of your idea

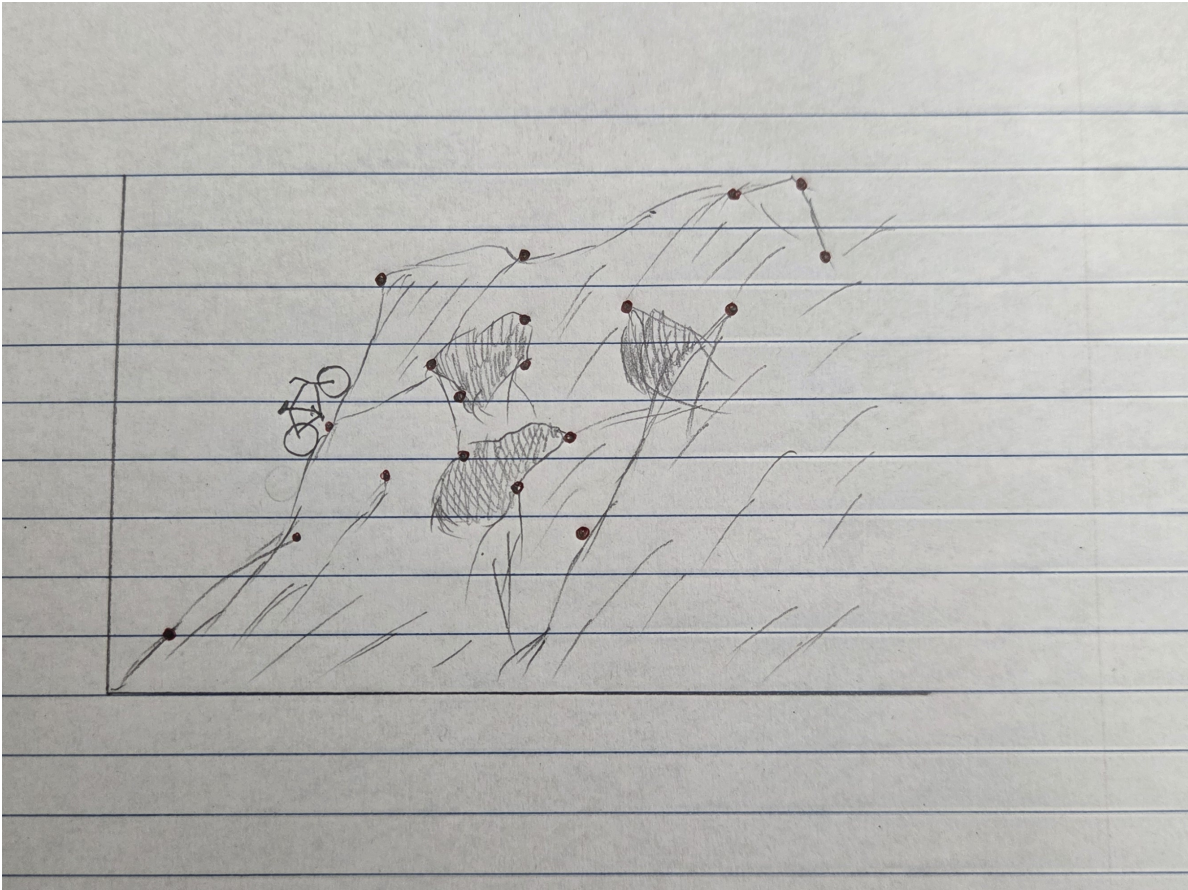


Figure 1: Sketch of idea

c. Make a draft of your visualization

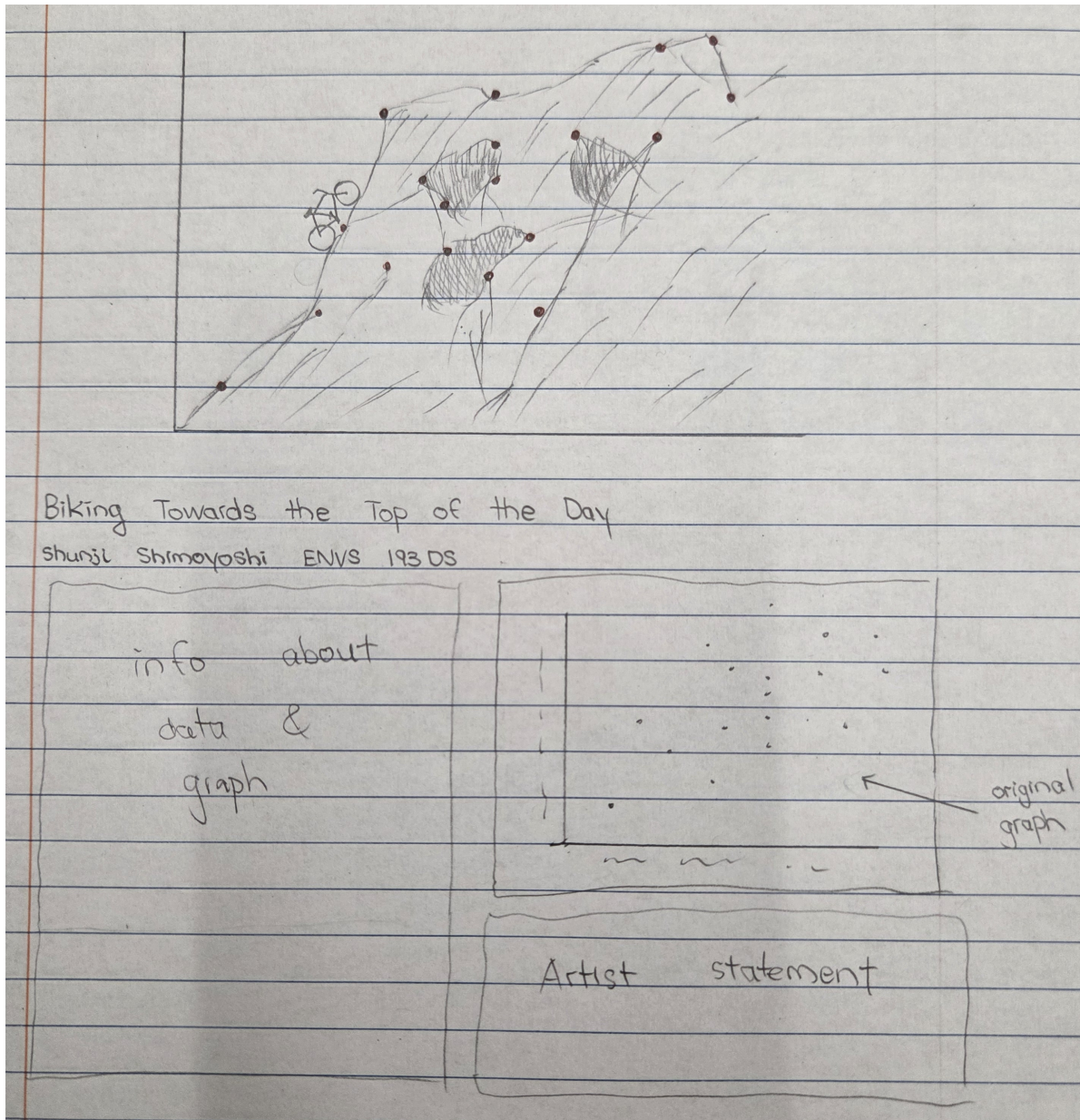


Figure 2: Draft of visualization

d. Write an artist statement

This piece is a visualization of my biking habits in spring quarter. While trying to ignore my packed schedule, I was thinking and became interested in seeing whether or not this had an effect on the amount of bike travel I did in a day to get where I need to be. I was inspired by Jill Pelto's techniques for this piece, as I thought her blending of art and data representation was interesting and beautiful. This digital art drawing depicts the individual data points as the rocky, chaparral-covered slopes of the Santa Ynez Mountains right above UCSB's campus. I collected the data over the course of around a month, while the digital drawing was done on Photoshop.

e. Prep your materials to share in class

Google slides presentation with my affective data visualization plan can be viewed [here](#).

Problem 4. Statistical critique

a. Revisit and summarize

The authors tested the hypothesis that California abalone species are negatively affected by warming ocean temperatures.

- Test used: ANOVA (Analysis of Variance)
- Response Variable(s): Abalone growth, reproductive output, disease, mortality
- Predictor Variable(s): Temperature, food quantity, food quality

The authors would interpret a significant result ($p < 0.05$) as there is a difference in average abalone growth, reproduction, disease, and mortality between the two study species, *H. rufescens* and *H. fulgens*.

TABLE 1. *F* values and probabilities from three-factor ANOVAs with equal replication, testing the effects of varying temperature, food quantity, and food quality on abalone growth, reproductive output, disease, and mortality.

Source of variation	df	Growth						Reproduction (GBI)		
		Length			Mass			MS	F	P
		MS	F	P	MS	F	P			
<i>Haliotis rufescens</i>										
Temperature	2	0.05	181.87	0.00	1.80	276.15	0.00	0.22	3.87	0.03
Quantity	2	0.08	272.42	0.00	1.40	215.86	0.00	1.86	32.51	0.00
Quality	2	0.02	79.22	0.00	0.51	79.08	0.00	0.50	8.74	0.00
Temperature : Quantity	4	0.00	4.16	0.01	0.02	3.71	0.01	0.05	0.90	0.47
Temperature : Quality	4	0.00	4.76	0.00	0.03	5.30	0.00	0.03	0.55	0.70
Quantity : Quality	4	0.00	10.22	0.00	0.06	9.55	0.00	0.29	5.14	0.00
Temperature : Quantity : Quality	8	0.00	0.34	0.94	0.01	1.06	0.41	0.03	0.59	0.78
Residuals	54	0.00			0.01			0.06		
<i>H. fulgens</i>										
Temperature	2	0.00	11.34	0.00	0.01	2.41	0.10	0.84	8.05	0.00
Quantity	2	0.12	690.91	0.00	1.57	474.76	0.00	7.32	70.11	0.00
Quality	2	0.02	133.68	0.00	0.29	88.96	0.00	0.36	3.45	0.04
Temperature : Quantity	4	0.00	9.10	0.00	0.05	14.36	0.00	0.52	4.99	0.00
Temperature : Quality	4	0.00	1.46	0.23	0.00	0.85	0.50	0.13	1.24	0.30
Quantity : Quality	4	0.00	22.46	0.00	0.03	10.10	0.00	0.05	0.47	0.76
Temperature : Quantity : Quality	8	0.00	0.82	0.59	0.00	1.08	0.39	0.08	0.78	0.62
Residuals	54	0.00			0.00			0.10		

Notes: Growth was evaluated through standardized changes in animal mass and shell length [(initial – final value)/initial value]. Reproductive output was evaluated from gonad bulk indices (GBI). Disease was evaluated by the burden of withering syndrome, Rickettsiales-like prokaryote (WS-RLP) and scores of morphological changes associated with withering syndrome (digestive gland atrophy [DGA] and foot muscle degeneration [FM-DG]), and accumulated mortalities. All variables were derived from cell means since that was the experimental unit and not individual abalone (see *Methods* for details). A posteriori tests were run on those treatments with significant effects (see Appendix).

TABLE 1. Extended.

Disease and morbidity											
WS-RLP			DGA			FM-Deg			Mortality		
MS	<i>F</i>	<i>P</i>	MS	<i>F</i>	<i>P</i>	MS	<i>F</i>	<i>P</i>	MS	<i>F</i>	<i>P</i>
4.74	19.64	0.00	2.64	25.32	0.00	1.80	9.48	0.00	0.38	41.11	0.00
0.34	1.40	0.25	1.78	17.07	0.00	2.08	10.90	0.00	0.04	3.97	0.02
0.07	0.28	0.76	0.40	3.84	0.03	0.98	5.13	0.01	0.00	0.08	0.92
0.19	0.79	0.54	0.32	3.05	0.02	0.15	0.81	0.53	0.03	3.49	0.01
0.10	0.41	0.80	0.15	1.43	0.24	0.05	0.25	0.91	0.00	0.20	0.94
0.31	1.29	0.28	0.02	0.23	0.92	0.24	1.24	0.30	0.00	0.16	0.96
0.35	1.45	0.20	0.12	1.14	0.35	0.08	0.43	0.90	0.00	0.34	0.94
0.24			0.10			0.19			0.01		
0.02	0.86	0.43	0.00	1.00	0.37	0.08	1.19	0.31	0.00	2.00	0.15
0.00	0.21	0.81	0.00	1.00	0.37	0.05	0.81	0.45	0.00	0.50	0.61
0.02	0.86	0.43	0.00	1.00	0.37	0.26	3.94	0.03	0.00	0.50	0.61
0.02	1.07	0.38	0.00	1.00	0.42	0.03	0.50	0.74	0.00	0.50	0.74
0.03	1.71	0.16	0.00	1.00	0.42	0.07	1.09	0.37	0.00	0.50	0.74
0.02	1.07	0.38	0.00	1.00	0.42	0.04	0.63	0.65	0.00	1.25	0.30
0.01	0.64	0.74	0.00	1.00	0.45	0.05	0.69	0.70	0.00	1.25	0.29
0.02			0.00			0.07			0.00		

b. Visual clarity

The information presented in the table represents the ANOVA test conducted by the authors fairly well. Important information for each response variable (ex. abalone growth, reproductive output, etc.) is clearly shown and organized, such as the degrees of freedom, F-statistic, mean squares, and p-value.

c. Aesthetic clarity

Given the numerous response variables for both species of abalone represented in the table, the authors handled the aesthetic clarity very well. Acronyms for variables, such as GBI (gonad bulk indices) are defined in the notes section below the table to help with better understanding. Additionally, grouping together variables such as “length” and “mass” under the larger header “growth” helps the reader understand that these two measures together represent abalone growth in this study. The same is done for “disease and mortality”, which groups together all of the measures of abalone diseases.

d. Recommendations

I would recommend that the authors add the effect size as an additional piece of information to this table. Doing so would further support their findings as they could show how large the difference is between the two abalone species for each response variable.